



Chapter Seven

Memory Organization

Outline

- Memory Hierarchy

- Main Memory

- Cache memory

 - Mapping Functions

 - Direct Mapping

 - Associative Mapping

 - Set Associative Mapping

- External Memory

 - Magnetic Disks

 - RAID Technology

 - Optical disks

Magnetic Tape

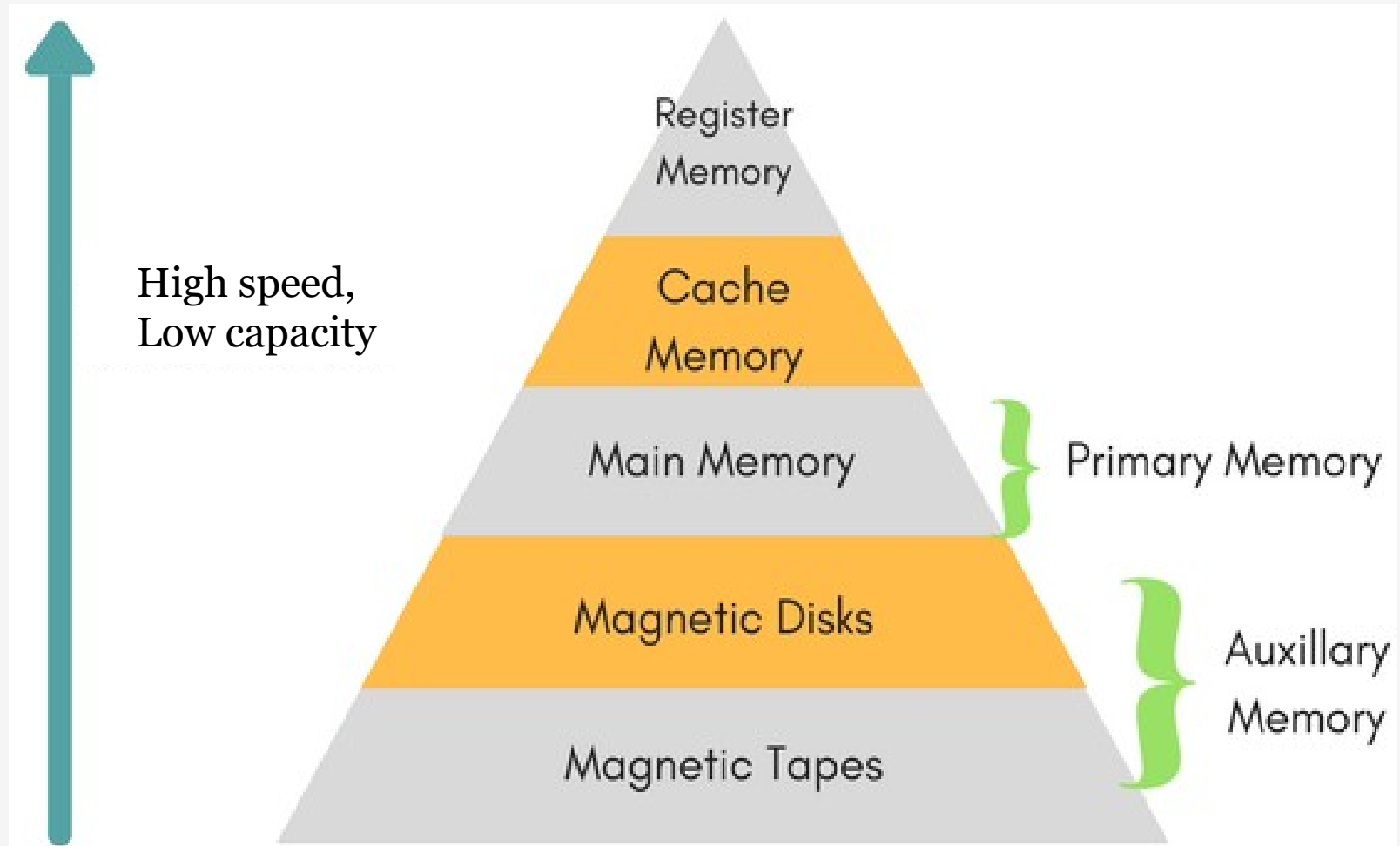
What is Memory Hierarchy?

- ❖ The memory in a computer can be divided into five hierarchies based on the speed as well as use.
- ❖ The processor can move from one level to another based on its requirements.
- ❖ The five hierarchies in the memory are registers, cache, main memory, magnetic discs, and magnetic tapes.
- ❖ The first three hierarchies are volatile memories
- ❖ Whereas the last two hierarchies are not volatile
- ❖ In general, the storage of memory can be classified into two categories such as volatile as well as non-volatile

Memory Hierarchy in Computer Architecture

- ❖ The **memory hierarchy design** in a computer system mainly includes different storage devices.
- ❖ Most of the computers were inbuilt with extra storage to run more powerfully beyond the main memory capacity.
- ❖ The following **memory hierarchy diagram** is a hierarchical pyramid for computer memory.

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The designing of the memory hierarchy is divided into two types

Primary Memory (Internal)

- ❖ The primary memory is also known as internal memory, and this is accessible by the processor straightly. This memory includes main, cache, as well as CPU registers.

Secondary memory(External)

- ❖ The secondary memory is also known as external memory, and this is accessible by the processor through an input/output module. This memory includes an optical disk, magnetic disk, and magnetic tape.

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Cache Memory

- ❖ Cache memory, also called CPU memory, is random access memory (RAM) that a computer microprocessor can access *more quickly than it can access regular RAM*.
- ❖ This memory is typically integrated directly with the CPU chip or placed on a separate chip that has a separate bus interconnect with the CPU.
- ❖ The basic purpose of cache memory is to store program instructions that are frequently re-referenced by software during operation.
- ❖ Fast access to these instructions increases the overall speed of the software program.

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Cache memory levels

- ❖ Cache memory is fast and expensive. Traditionally, it is categorized as "levels" that describe its *closeness and accessibility to* the microprocessor:
- ❖ **Level 1 (L1)** cache is extremely fast but relatively small, and is usually embedded in the processor chip (CPU).
- ❖ **Level 2 (L2)** cache is often more capacious than L1;
 - ✓ it may be located on the CPU or on a separate chip or
- ❖ **Level 3 (L3)** cache is typically specialized memory that works to improve the performance of L1 and L2.
 - ✓ It can be significantly slower than L1 or L2, but is usually double the speed of RAM.

Cache entries

- ❖ Data is transferred between memory and cache in blocks of fixed size, called cache lines.
- ❖ When a cache line is copied from memory into the cache, a cache entry is created.
- ❖ The cache entry will include the copied data as well as the requested memory location (now called a tag).
- ❖ When the processor needs to read or write a location in main memory, it first checks for a corresponding entry in the cache.
- ❖ The cache checks for the contents of the requested memory location in any cache lines that might contain that address.

Replacement policies

- ❖ In order to make room for the new entry on a cache miss, the cache may have to evict one of the existing entries.
- ❖ The heuristic that it uses to choose the entry to evict is **called the replacement policy..**
- ❖ One popular replacement policy, least-recently used (LRU), replaces the least recently accessed entry.

CPU stalls

- ❖ The time taken to fetch one cache line from memory (read latency) matters because the CPU will run out of things to do while waiting for the cache line.

When a CPU reaches this state, it is called a stall.

- ❖ Various techniques have been employed to keep the CPU busy during this time.
 - ✓ Out-of-order CPUs (the Pentium Pro and later Intel designs, for example) attempt to execute independent instructions after the instruction that is waiting for the cache miss data.
 - ✓ simultaneous multithreading (SMT) which allows an alternate thread to use the CPU core while a first thread waits for data to come from main memory.

Types of External Memory

Magnetic Disk

- RAID
- Removable

Optical

- CD-ROM
- CD-Recordable (CD-R)
- CD-R/W
- DVD

Magnetic Tape

Magnetic Disk Memory

- ❖ **Magnetic Disk** is type of secondary memory which is a flat disc covered with magnetic coating to hold information.
- ❖ It is used to store various programs and files.
- ❖ The polarized information in one direction is represented by 1, and vice versa. The direction is indicated by 0.
- ❖ Magnetic disk are less expensive than RAM and can store large amounts of data, but data access rate is slower than main memory because of secondary memory.
- ❖ Data can be modified or can be deleted easily in the magnetic disk memory.

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Advantages

- These are economical memory
- It can store large amounts of data.
- It has better data transfer rate than magnetic tapes.
- It has less prone to corruption of data as compared to tapes.

Disadvantages

- ❖ It need clean and dust free environment to store.
- ❖ These are not suitable for sequential access.

RAID (redundant array of independent disks)

What is RAID?

- ❖ RAID is a way of storing the same data in different places on multiple hard disks or solid-state drives (SSDs)
- ❖ There are different RAID levels, however, and not all have the goal of providing redundancy.

How RAID works

- ❖ RAID works by placing data on multiple disks and allowing input/output (I/O) operations to overlap in a balanced way, improving performance.
- ❖ Because using multiple disks increases the mean time between failures, storing data redundantly also increases fault tolerance.
- ❖ RAID employs the techniques of ***disk mirroring or disk striping***.
 - **Mirroring** will copy identical data onto more than one drive.
 - **Striping partitions** help spread data over multiple disk drives.
 - ✓ Each drive's storage space is divided into units ranging from a sector of 512 bytes up to several megabytes.
 - ✓ The stripes of all the disks are interleaved and addressed in order.

RAID controller

- ❖ A RAID controller is a device used to manage hard disk drives in a storage array.
- ❖ It can be used as a level of abstraction between the OS and the physical disks.
- ❖ Using a RAID controller can improve performance and help protect data in case of a crash.

RAID levels

- ❖ RAID devices use different versions, called levels.

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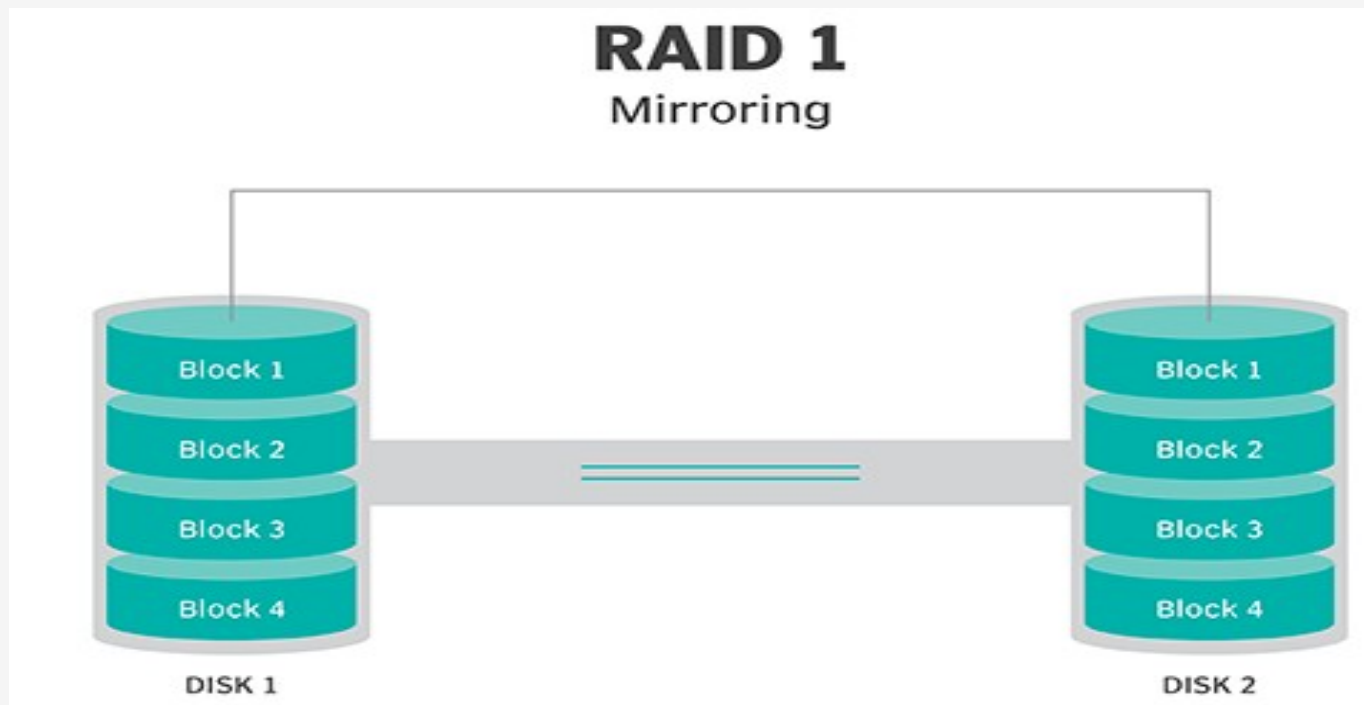
RAID 0. This configuration has striping but no redundancy of data. It offers the best performance, but it does not provide fault tolerance.



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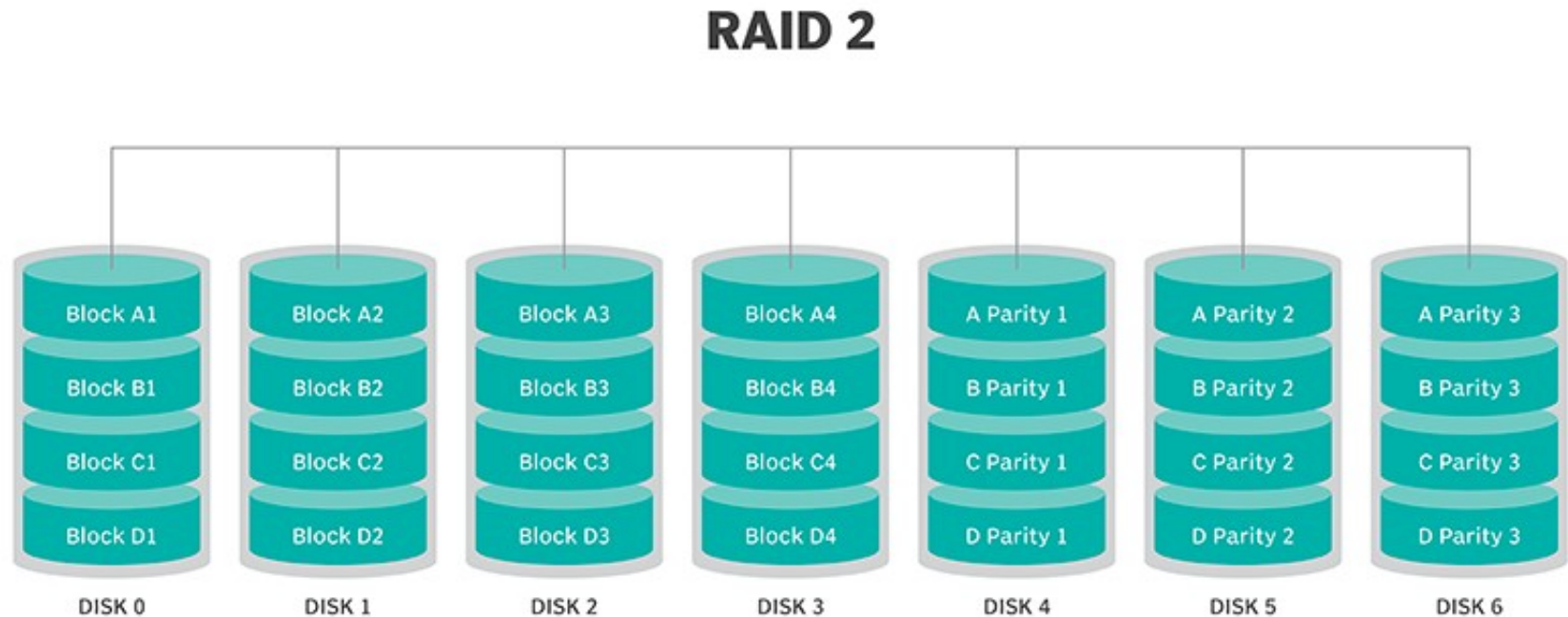
RAID 1. Also known as disk mirroring, this configuration consists of at least two drives that duplicate the storage of data.

- ✓ There is no striping. Read performance is improved, since either disk can be read at the same time. Write performance is the same as for single disk storage.



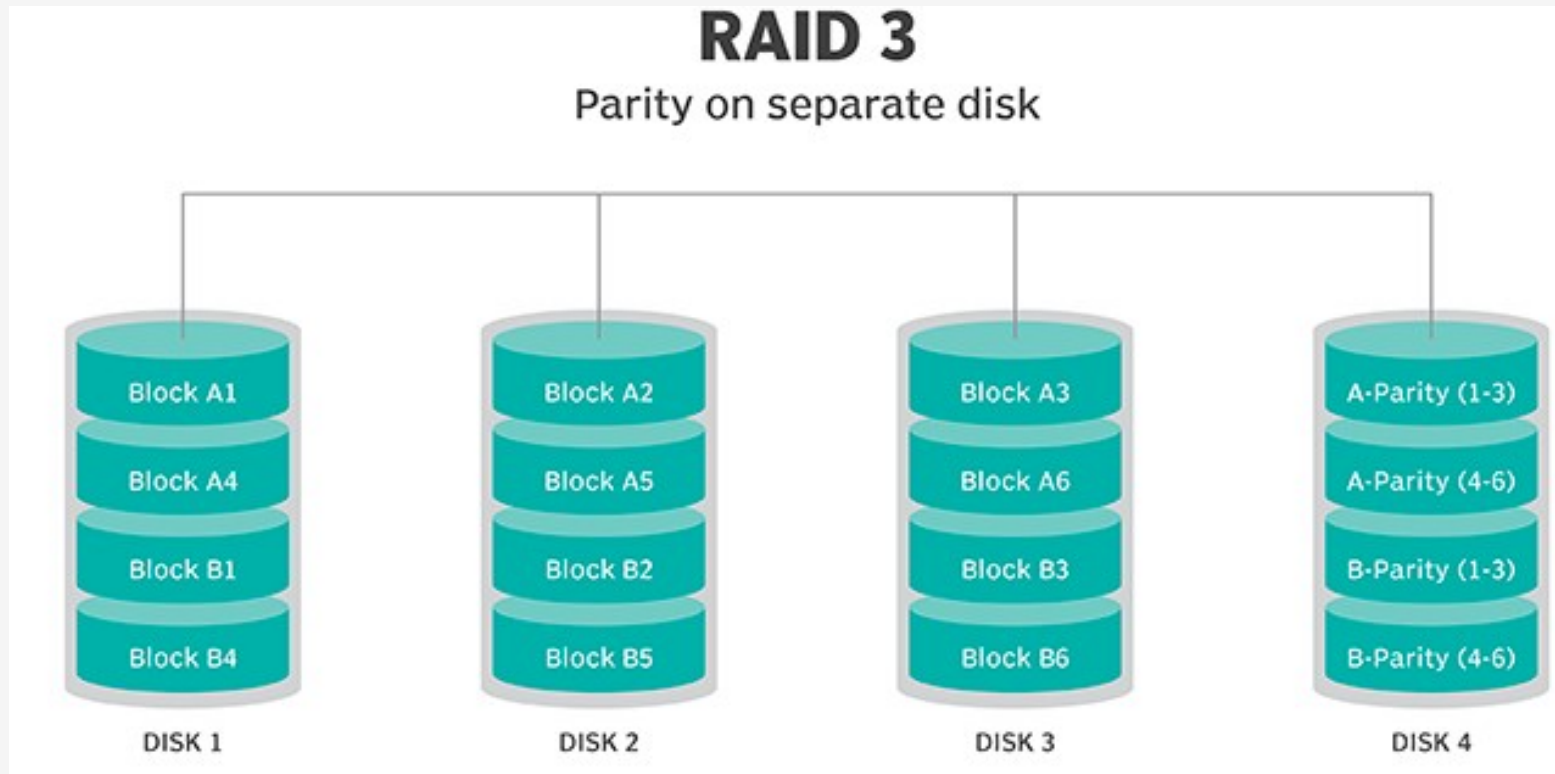
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RAID 2. This configuration uses striping across disks, with some disks storing error checking and correcting (ECC) information. RAID 2 also uses a dedicated Hamming code parity, a linear form of ECC.



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RAID 3. This technique uses striping and dedicates one drive to storing parity information. The embedded ECC information is used to detect errors



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- ❖ **compact disc read-only memory (CD-ROM)** :- This type of CD doesn't allow users to erase nor write over whatever is in the disc.
- ❖ **CD-Recordable (CD-R)** :- A CD like this gives users the ability to write on the CD only once but has an unlimited number of reads available.
- ❖ **CD Re-Writable (CD-RW)** — With a CD like this, you can erase and write over it for up to 1000 times.
- ❖ **Digital Versatile Discs (DVDs)** :-are much like CDs in that it also uses laser light to store and retrieve data. This particular way of storing or retrieving data is characteristic of optical storage devices.
- ❖ **Magnetic tape** :-is one of the oldest technologies for electronic data storage. While tape has largely been displaced..

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